Correlators of boosted resonances

Simon Rothman

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1 Motivation

Measuring EECs of vector bosons is highly appealing. Potential applications include:

- Search for boosted narrow resonances decaying hadronically (as a bump hunt in the EEC spectrum). (Are we sensitive to Higgs??).
- Calibration/proof-of-principle for EEC top mass measurement
- Measurement of transverse/longitudinal polarization fraction of high-pT Z and W
- Measurement of the full production spin-density matrix for high-pT Z and W bosons

2 Background

The two-point energy correlator (EEC) of a boosted, hadronically decaying boson looks very different from that of a QCD jet, whose collinear spectrum is smooth and approximately scale invariant [1]. A boson of boost $\gamma = E_V/m_V$ instead produces a decay peak at $z' \simeq 1/\gamma^2$, with $z' = (1 - \cos\theta')/2$, which at central rapidity corresponds to $\Delta R \simeq 2m_V/p_T$ [1]. The width of the peak is set mainly by QCD radiation from the quarks rather than by Γ_V/m_V, and well below the peak the spectrum follows that of a quark jet [2].

2.1 Polarization and production

In the narrow-width approximation, the correlator factorizes into a production density matrix and a decay matrix [1, 2]. The opening-angle spectrum depends on production only through the longitudinal fraction ρ_{00} [1]. Using NNLO production for $pp \to V + X$ and decay correlators at NNLO with NNLL collinear and N^{4}LL back-to-back resummation, Gao et al. find that the peak height at $\gamma = 4$ differs by about 70% between purely longitudinal and purely transverse bosons. For $p_T > 100$ GeV at $y_V = 0$, the polarized spectrum is suppressed by up to about 10% across the peak relative to an unpolarized treatment, because high-p_T production is dominantly transverse [1]. The off-diagonal entries require measuring the detector orientation relative to the production plane [1]: $|\Delta h| = 2$ interference appears in the azimuthal angle, while $|\Delta h| = 1$ interference requires the azimuth and the angular separation jointly [2]. One-point energy and charge correlators together determine all eight real parameters of the density matrix [1]. Two experimental cautions come from Ref. [2]: boosts along the LHC beam axis modify the azimuthal distributions, an effect expected to be mild but not yet assessed, and in a lepton-collider study D_2 selections preferentially reject transverse bosons, homogenizing the longitudinal and transverse one-point spectra.

Two neighboring results extend the toolkit. The one-point correlator of a vector current is fixed by a single parameter $a_{\mathcal{E}}$, bounded by unitarity to $-1/2 \le a_{\mathcal{E}} \le 1$; its N^{3}LO value at $q^2 = m_Z^2$ is -0.42, in good agreement with values extracted indirectly from LEP data [3]. Flavor flux correlators generalize the program to flavor quantum numbers, with hadronization corrections to the b and c cases suppressed by $\Lambda_{\mathrm{QCD}}/m_Q$ [4].

2.2 Searches

The same peak motivates resonance searches. For a narrow resonance coupled linearly to a QCD operator, the peak position encodes the mass and the spectral shape encodes the spin and polarization, within a space of shapes bounded by unitarity and energy positivity [5]. A projection for a light hadrophilic Z' in $pp \to Z'\gamma$, with statistical uncertainties only, finds that the inclusive EEC reaches sensitivity comparable to a recast of a CMS soft-drop mass search [5]. A separate recast of the CMS dijet correlator measurement also gives competitive limits, although its signal is at particle level and detector effects may degrade the reach [5]. In a Monte Carlo study of jets with $p_T > 1$ TeV recoiling against $Z \to \ell\ell$, hadronic W/Z decays appear as a local excess of about 7% even though they make up only about 1% of the jets, with no reconstruction or tagging; that selection is better suited to the HL-LHC than to existing datasets [1]. The peak position also calibrates the angular scale, making m_V a standard candle inside jets, including the polarized W inside boosted top jets [1].

3 Questions and Goals

In the short term, we want to understand how we can use these ideas to do a CMS measurement. A few important lines of questioning:

"Untagged" resonance search

- What sensitivity might be expect in an "untagged"-style search for a narrow resonance in Run 3 data?
- Which dataset(s) would we want to use?
  - Z+jet (tagged by Z$\to \ell\ell$)? Potential advantage: polarization of the other Z from leptonic decay?
  - Photon+jet (tagged by photon)? Potential advantage: larger cross-section. Photon cannot be longitudinally polarized, I assume this has some implications for the polarization of the Z?
  - Dijet (with exactly two jets and momentum balance)? Potential advantage: large cross-section, but also large QCD bacground.
  - QCD multijet (without momentum balance)? Potential advantage: largest cross-section, but also largest QCD background.
  - Something else?
- What do various SM processes look like?
  - Can we separate W and Z at all?
  - Can we see the Higgs? If not, can we set limits? Would they be competitive?
  - What does the top look like? Do we learn anything interesting?
- Do we get any extra sensitivity from observables beyond two-point correlator? E.g,. E3C, E1C, etc. Including non-standard formulations thereof? The paper mentions E^2E^2 for example. Or with non-standard angles as in the one-point from the paper...
- Would we get boosted sensitivity from some additional cuts?
- Is heavy flavor interesting? Can we see heavy mesons?
- Is the background really as clean as we hope it is?

- How bad is the jet energy uncertainty? Is that the dominant systematic? What other systematics could be important?
- Would our limits actually be competitive?

Polarization measurements

- How good of a job do we need to do at tagging? Can we do it untagged? Is there any way to do good tagging that doesn't sculpt the distributions to shit?
- Is there a useful shape difference between longitudinal and transverse bosons? Or is it dominantly the peak height?
- Are the one-point correlators actually practical? What does the background do to them?
- Would the three-point correlator be useful/interesting? Or some related things, maybe non-standard ones...
- Is there an interesting bell inequality story here? Can we see interference effects?
- Again, which dataset(s) would we want to use?
- Charge correlators? Are they useful? How badly does hadronization etc. ruin the picture?
- How large are the spin analyzer coefficients?
- What is the other literature? Are we competitive?

4 Tasks and Timeline

Break down the work into manageable tasks with deadlines.

1. Task 1
2. Task 2
3. Task 3

5 Resources

List required resources, dependencies, or tools.

6 Risks and Mitigations

Identify potential risks and how they will be addressed.

7 Success Criteria

Define what constitutes successful completion.

References

[1] Anjie Gao, Kyle Lee, and Xiaoyuan Zhang. *Precision Studies of Boosted Polarized Vector Bosons with Energy Correlators*. 2026. arXiv: 2610.38688 [hep-ph].

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[3] Marc Riembau and Minho Son. "Flow between Extremal One-Point Energy Correlators in QCD". In: *Physical Review Letters* 136.10 (2026), p. 101901. DOI: 10.1103/lc5m-kwgv. arXiv: 2509.02669 [hep-ph].

[4] Kyle Lee and Ian Moult. *Flavor Flux Correlators, Forward-Backward Asymmetries, and Anomalies*. 2026. arXiv: 2609.13366 [hep-ph].

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